

COLLEGE OF DEFENCE MANAGEMENT · DATATHON - 2026

Theme 6.1: Contested and Congested – Growth of Satellites in Outer Space

PROJECT HIGH GROUND

Contested and Congested: the Crowding of Orbit, and What It Means for India and the Indian Armed Forces

Commander's Edition

How crowded orbit has become since 2014, who owns it, how dangerous it is, who is building military eyes in it, what nature adds, where it is heading by 2030, and what India and its Armed Forces should do. Told as one story, in plain words.

Submitted by

[Rank] ARJUN JAYADEV

[Service No] · [Unit / Formation]

September 2026

Companion files: Power BI dashboard (.pbix), analysis code and data

THE STUDY AT A GLANCE

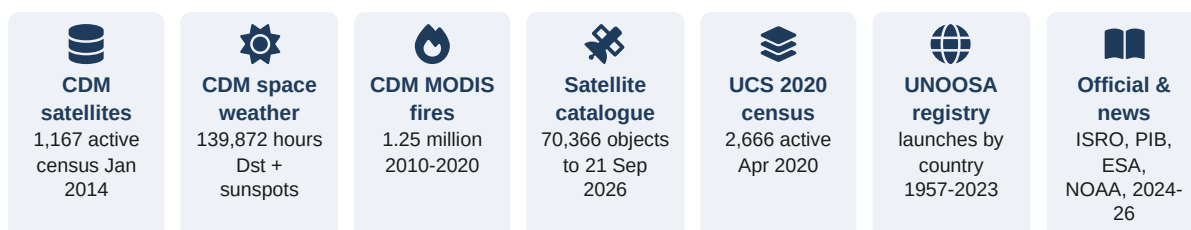
PROJECT HIGH GROUND · Contested and Congested: the Crowding of Orbit

CDM Datathon-2026 · Theme 6.1: Growth of Satellites in Outer Space · the study on one page

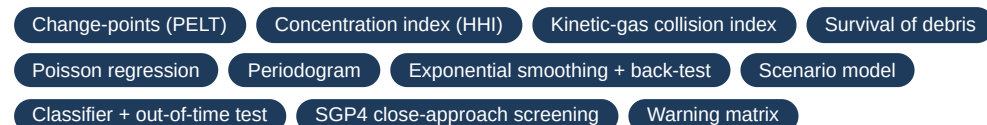
THE STORY IN SEVEN STEPS



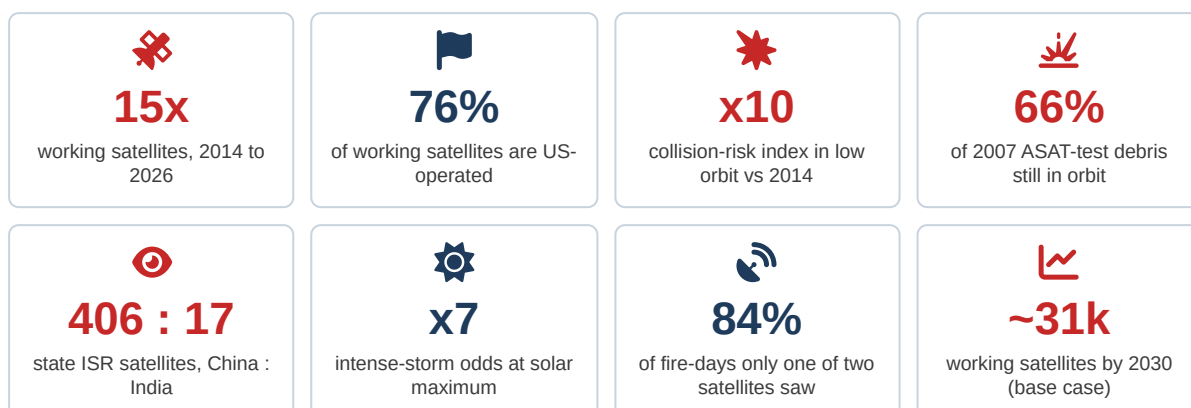
DATA FUSED



METHODS



WHAT THE DATA SAYS



Bottom line: orbit is crowded, concentrated and contested. India depends on it more each year while holding a shrinking share of it. Build capacity and resilience, and watch the orbit weekly.

EXECUTIVE SUMMARY

BOTTOM LINE UP FRONT. Orbit has become **crowded, concentrated and contested** faster than any plan assumed. Working satellites rose from 1,167 to 17,062 in twelve years; collision risk in low orbit is about **10 times** the 2014 level; and China now operates about **406** state and military imaging and intelligence satellites to India's **17**. India's share of working satellites has fallen to 0.5%.

Situation

The CDM datasets give a census of 1,167 active satellites (Jan 2014), 140,000 hours of space-weather records and 1.25 million satellite fire detections. Open sources extend the picture to September 2026: the full public satellite catalogue (70,366 objects), the UCS 2020 census and UN launch registrations.

Assessment

1. **Crowded.** Payloads launched a year rose from 199 (2013) to 4,480 (2025). Growth changed gear around 2021.
2. **Concentrated.** The USA operates 76% of working satellites (43% in 2014); one company about 65%.
3. **Risky.** A live crowd of 9,964 objects sits at 450-500 km; a dead crowd of debris at 750-900 km. Debris from China's 2007 weapon test is still 66% in orbit; India's 2019 test debris has all re-entered.
4. **Contested.** China has launched about 43 state intelligence satellites a year since 2023; India under one. SBS-III's 52 satellites close only about 13% of today's gap.
5. **Exposed to nature.** At high solar activity the odds of an intense magnetic storm in a month rise from 6% to 39%.
6. **Close calls daily.** A one-day screening of current orbits found 18,355 passes under 5 km; 10 Indian satellites, including EMISAT, passed within 5 km of weapon-test or collision debris.
7. **Worth it.** Two satellites saw 84% of fire-days that one alone would have missed: numbers buy coverage.

Deduction

India's security now depends on space, while its share of space shrinks and the environment grows more dangerous. Capacity, resilience and awareness of the space picture, not single prestige satellites, decide who keeps the high ground.

Recommendation

- Stand up **ORBITWATCH** under the Defence Space Agency: a weekly space-situation picture and warning matrix from open and national data.
- Accelerate SBS-III and buy commercial capacity to reach **persistent** coverage of the northern borders and the Indian Ocean.
- Make every new military satellite **manoeuvrable and storm-hardened**, and budget fuel for collision avoidance.

Decision sought

Approve a **four-month ORBITWATCH proof of concept** at the Defence Space Agency using the open catalogue and the pipeline already built. Continue only if it flags at least one actionable conjunction or threat event.

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Each chapter opens with **The story so far** and closes with **So what**; every chart carries **What this shows**. A commander can read the Executive Summary and the coloured boxes alone in about ten minutes.

1. THE QUESTION AND HOW WE ANSWERED IT

PROLOGUE - THE QUESTION

THE STORY SO FAR

In May 2025, during Operation Sindoor, ten Indian satellites worked round the clock to see, talk and navigate for the forces. Space is no longer a support service; it is where modern operations begin. This booklet asks what has happened to that space in the last twelve years, and what India and its Armed Forces should do about it.

Theme 6.1 asks participants to derive insights into the proliferation of satellites and to predict the future trajectory. This study turns that into five questions a commander would ask: **How crowded is orbit? Whose satellites are they? How dangerous has it become? Who is building military eyes in orbit? And what should India do?**

GEO 35,786 km	Geostationary belt: communications, early warning, weather. The arc used by India (40-110 E) is among the most crowded on the belt.	162 satellites 40-110 E
MEO ~20,000 km	Navigation constellations (GPS, GLONASS, Galileo, BeiDou). India's NavIC sits in GEO/inclined orbits.	4 global GNSS systems
LEO 1,050-1,200 km	Coming: China's Guowang and Qianfan shells (about 27,000 planned). Debris here stays for centuries.	~470 already launched
LEO 750-900 km	THE DEAD CROWD: 82% debris, much of it from the 2007 Chinese ASAT test and the 2009 collision.	4,583 objects
LEO 450-500 km	THE LIVE CROWD: 99% working satellites, mostly Starlink. Up from 425 objects in 2014.	9,964 objects
LEO < 450 km	Very low orbits: short-lived; where India's 2019 ASAT test was conducted, so its debris re-entered fast.	0 Shakti debris left

Figure 1.1 : Orbit at a glance: where the crowds are, from 300 km to 36,000 km (numbers from this study)



Figure 1.2 : How the question was answered

The three CDM datasets give a census of **1,167 active satellites** (January 2014), **139,872 hours** of space-weather records, and **1.25 million** fire detections by two NASA satellites. Open sources bring the story to September 2026: the full public satellite catalogue (70,366 objects), the UCS 2020 census and UN launch registrations. The glossary at Annex A explains each technique in plain words.

SO WHAT

Five questions, three CDM datasets, four open sources and one repeatable pipeline.

Next: How crowded has orbit become?

2. THE CROWD

PART I - THE CROWD

THE STORY SO FAR

The story begins with numbers. In January 2014, the date of the CDM satellite census, about 1,200 satellites were working. What happened next?

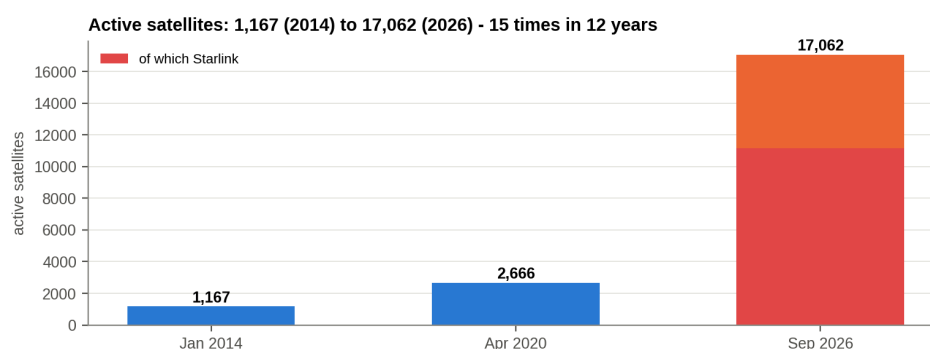


Figure 2.1 : Working satellites in three independent censuses: CDM (2014), UCS (2020), satellite catalogue (2026)

What this shows: Three independent counts of working satellites. Red is one company's constellation.

In January 2014 about **1,167** satellites were working. By April 2020 there were 2,666; by September 2026 **17,062**, a **15-fold** rise in twelve years. About 65% of today's working satellites belong to a single constellation, Starlink.

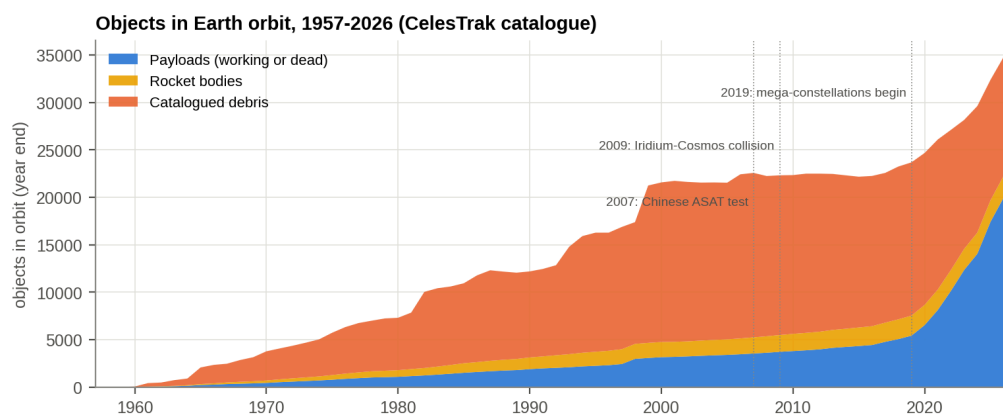
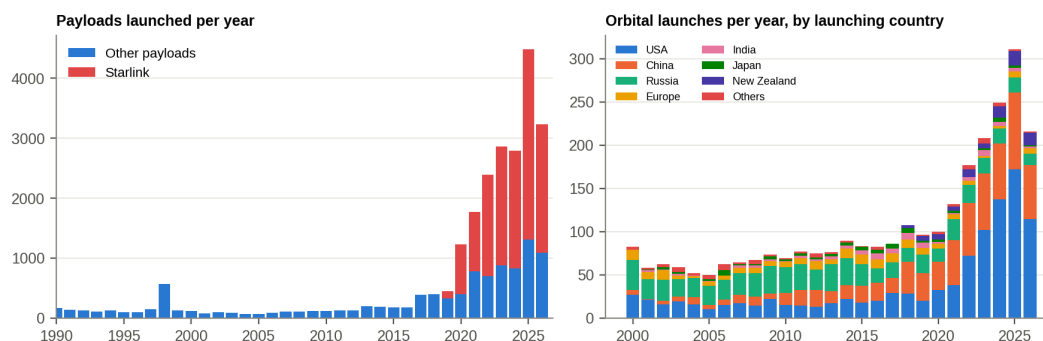


Figure 2.2 : Everything tracked in Earth orbit, 1957-2026: payloads, rocket bodies and debris

What this shows: Everything we track in orbit, year by year. Blue is satellites (working or dead), orange is debris. The blue wedge at the right is the mega-constellation era.

The launch rate changed gear after 2019 (2026 = to 21 Sep)**Figure 2.3 : Satellites launched and rocket launches per year**

What this shows: Left: satellites launched each year (red = Starlink). Right: rocket launches each year by country.

The launch rate changed gear. In 2013, 199 satellites were launched; in 2025, **4,480**. A change-point test places the break in **2021**: growth of 4% a year before the mega-constellations, **47% a year** after. Rocket launches rose from 76 (2013) to 311 (2025). UN registrations track the catalogue almost exactly (correlation 0.997), so the count is sound.

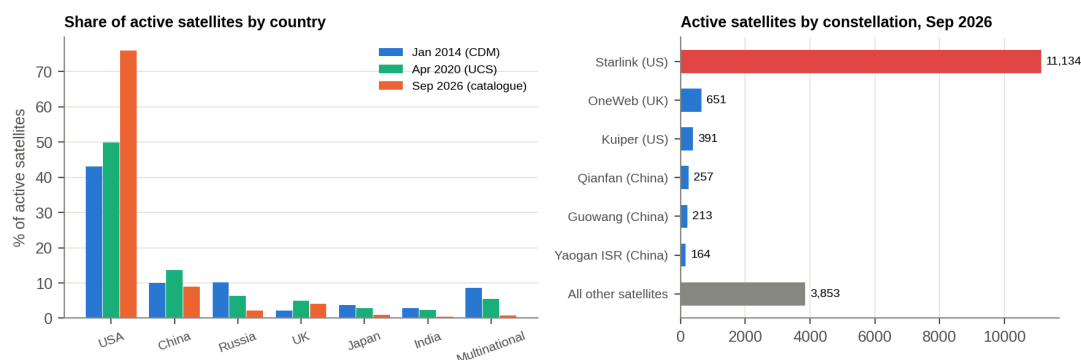
SO WHAT

Active satellites went from 1,167 to 17,062 in twelve years (15 times). Growth changed gear around 2021: 4% a year before, 47% after.

Next: Whose satellites are they?

3. WHO OWNS ORBIT**PART II - WHO OWNS ORBIT****THE STORY SO FAR**

The sky filled 15 times over. The next question is whose satellites these are, and what they are for.

Orbit is concentrating: one country, and one company, now hold most active satellites**Figure 3.1 : Share of working satellites by country, and the largest constellations today**

What this shows: Left: each country's share of working satellites in 2014, 2020 and 2026. Right: the largest constellations today.

In 2014 the USA operated 43% of working satellites; today **76%**. The concentration index (HHI) rose from 2,171 to **5,879**; anything above 2,500 counts as highly concentrated. China grew 13-fold in numbers (117 to 1,531) but held its share near 10%. **India grew from 33 to 78 satellites, but its share fell from 2.8% to 0.5%.**

Table 3.1 : Working satellites by country: 2014, 2020, 2026

Country	Jan 2014 (CDM)	Apr 2020 (UCS)	Sep 2026 (catalogue)	Jan 2014 (CDM) %	Sep 2026 (catalogue) %
USA	502	1,327	12,959	43	76
China	117	363	1,531	10	9
Russia	118	169	380	10.10	2.20
UK	25	130	698	2.10	4.10
Japan	43	78	149	3.70	0.90
India	33	61	78	2.80	0.50
Multinational	100	146	141	8.60	0.80
All others	229	392	1,124	19.60	6.60

What satellites are for, and how military they are

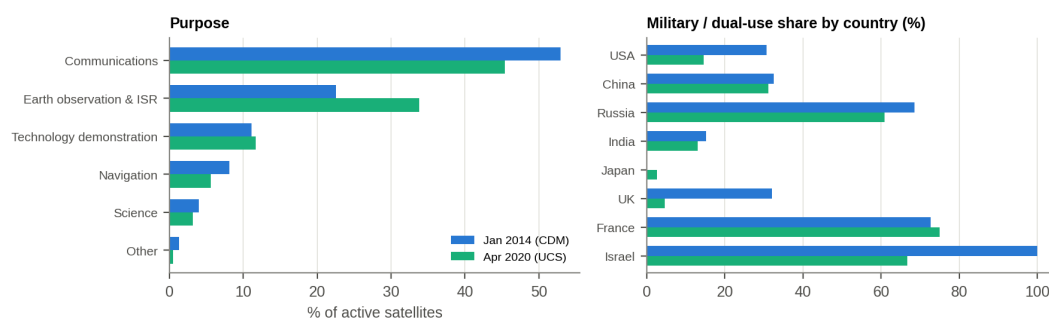


Figure 3.2 : What satellites are for, and how military each country's fleet is

What this shows: What satellites do, and what share of each country's fleet serves the military.

Commercial users rose from 36% of satellites (2014) to 54% (2020), while the military share fell from 28% to 18%. That is not demilitarisation: military satellites grew in number, and commercial constellations are now used in war (Ukraine has shown this). In 2014, 69% of Russian and 32% of Chinese satellites served the military, against 15% of India's. And 47% of the 2014 satellites were already past their design life: fleets must be renewed continuously just to stand still.

SO WHAT

One country operates 76% of working satellites and one company about 65%. India's share fell from 2.8% to 0.5%.

Next: Where is orbit crowded, and how risky is it?

4. WHERE IT IS CROWDED

PART III - WHERE IT IS CROWDED

THE STORY SO FAR

A few owners now hold most of the sky. But orbit is not one place: some altitudes are empty and some are packed. Where is the crowding, and how dangerous is it?

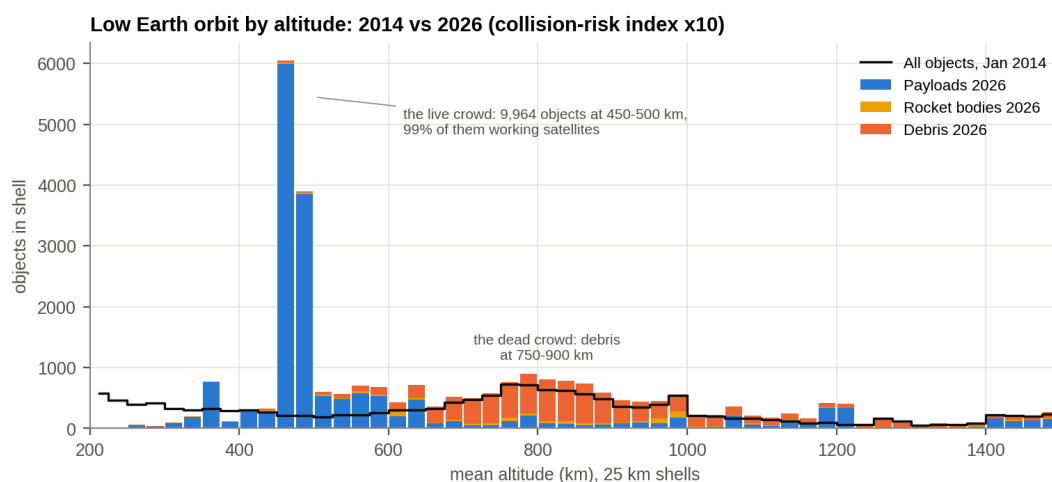


Figure 4.1 : Low Earth orbit by altitude, 2014 against 2026

What this shows: How many objects sit at each height in low orbit. The black line is 2014; bars are today. Tall blue bars are working satellites; orange is debris.

Orbit is crowded in two places. **The live crowd:** 9,964 objects at 450-500 km, 99% of them working satellites (there were 425 objects there in 2014). **The dead crowd:** 4,583 objects at 750-900 km, 82% of them debris. A kinetic-gas model, in which collisions grow with the square of the number of objects at each height, puts the **collision-risk index at about 10 times** the 2014 level.

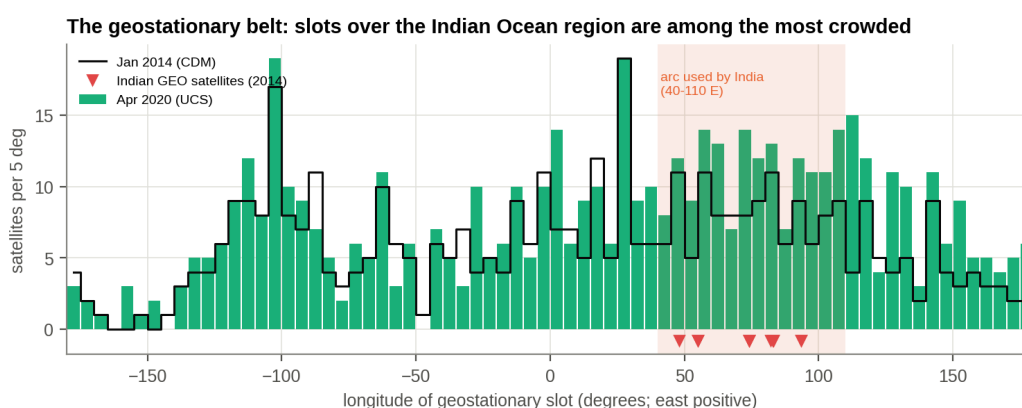


Figure 4.2 : The geostationary belt by longitude; the shaded arc is the one India uses

What this shows: Satellites parked above the equator, by longitude. The shaded band is the arc India uses; red markers are Indian satellites.

Higher up, the geostationary arc used by India (40-110 E) held **162** satellites in 2020, 29% of the whole belt, up from 115 in 2014. China (32) and the USA (33) each had more there than India (24). Slots and radio frequencies over India are a contested resource.

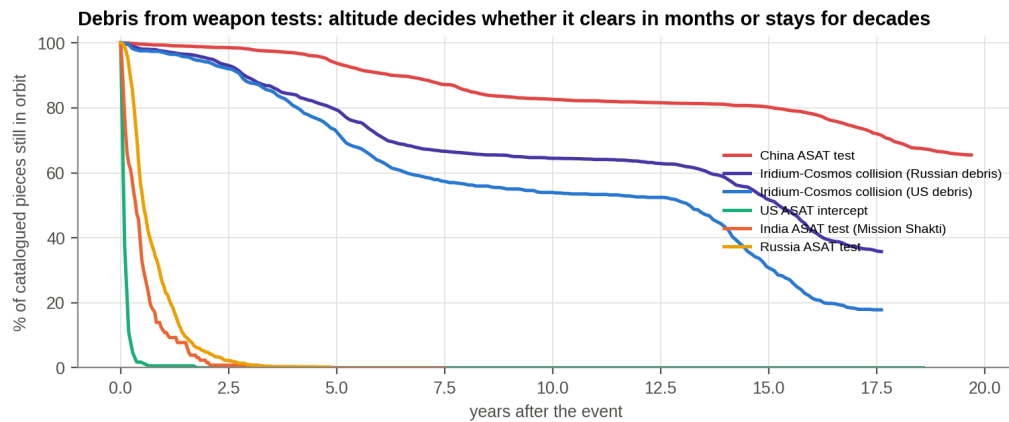


Figure 4.3 : Debris from weapon tests and the 2009 collision: share still in orbit, year by year

What this shows: Share of debris pieces from each weapon test or collision still in orbit, year by year afterwards.

Table 4.1 : Weapon tests and collisions: debris created and debris remaining

Event	Date	Pieces catalogued	Still in orbit (Sep 2026)	Still in orbit %
China ASAT test	Jan 2007	3,531	2,313	65.50
Iridium-Cosmos collision (Russian debris)	Feb 2009	1,714	613	35.80
Iridium-Cosmos collision (US debris)	Feb 2009	656	117	17.80
US ASAT intercept	Feb 2008	174	0	0
India ASAT test (Mission Shakti)	Mar 2019	129	0	0
Russia ASAT test	Nov 2021	1,806	4	0.20

Altitude decides the legacy. China's 2007 test at about 865 km made 3,531 tracked pieces; **66% are still in orbit** nearly twenty years later. India's 2019 test (Mission Shakti) was done at about 280 km: all 129 pieces have re-entered, most within half a year. Russia's 2021 test (1,806 pieces) has almost cleared. A responsible test is a low one; a high one poisons an orbit for a generation.

4.1 Close Calls: Screening Every Satellite for a Day

Close-approach screening from current orbital elements (SGP4)

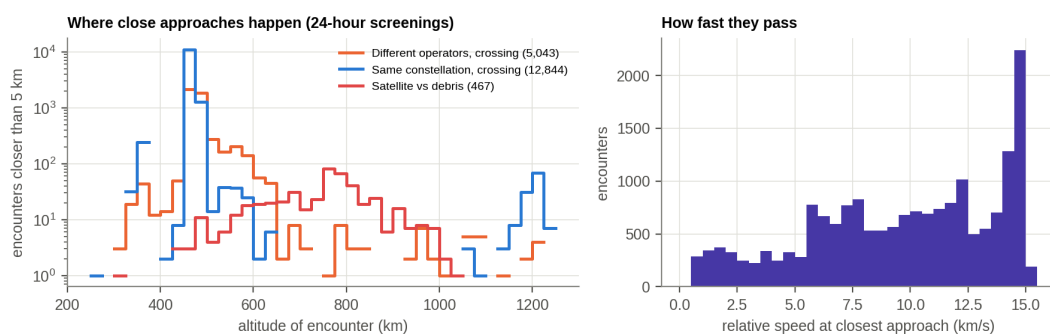


Figure 4.4 : Close approaches under 5 km found by screening current orbits with the SGP4 propagator

What this shows: Left: at what height satellites came within 5 km of each other or of debris in one day. Right: how fast they passed each other.

To see the congestion directly, every active satellite in low orbit (15,656) was propagated with SGP4, the standard model for published orbital elements, every 10 seconds for 24 hours, and every pair was

checked. A second screening pitted freshly tracked satellites against the four largest debris clouds. In **one day: 17,887 crossing encounters closer than 5 km** between satellites (385 under 1 km), 93% of them involving Starlink, typically at 10 km/s; and **468 encounters between satellites and weapon-test or collision debris** (20 under 1 km). The closest pass found was 47 m.

Table 4.2 : Indian satellites passing within 5 km of weapon-test and collision debris, 23-24 Sep 2026

Indian satellite	Debris object	Miss distance (km)	Relative speed (km/s)	Altitude (km)
SARAL	COSMOS 2251 DEB	1.44	4.73	789
SCATSAT 1	COSMOS 2251 DEB	2.71	14.23	660
HYSIS	FENGYUN 1C DEB	3.02	4.07	627
EMISAT	COSMOS 2251 DEB	3.72	14.53	743
ASTROSAT	COSMOS 2251 DEB	3.93	9.35	618
EOS-6 (OCEANSAT-3)	COSMOS 2251 DEB	3.94	10.39	745
SARAL	FENGYUN 1C DEB	4.01	1.82	789
SRMSAT	FENGYUN 1C DEB	4.25	10.34	855

10 Indian satellites had such a pass in a single day, including EMISAT (electronic intelligence), HySIS, SARAL and Oceansat-3, mostly against debris from the 2009 Iridium-Cosmos collision and China's 2007 test. Public elements are accurate to about a kilometre, so this is screening, not collision probability, but it is exactly the daily watch ORBITWATCH would run.

SO WHAT

Collision risk in low orbit is about 10 times the 2014 level, and a one-day screening found 18,355 close approaches under 5 km, 10 of them Indian satellites passing debris. High-altitude weapon debris lasts decades; India's low-altitude test cleaned itself up within a year.

Next: Who is building military eyes in orbit?

5. THE CONTEST

PART IV - THE CONTEST

THE STORY SO FAR

A crowded orbit is a hazard. A crowded orbit full of military eyes is a contest. Who is building the watchers?

The ISR race in orbit: China and the USA build by the hundred; India by the handful

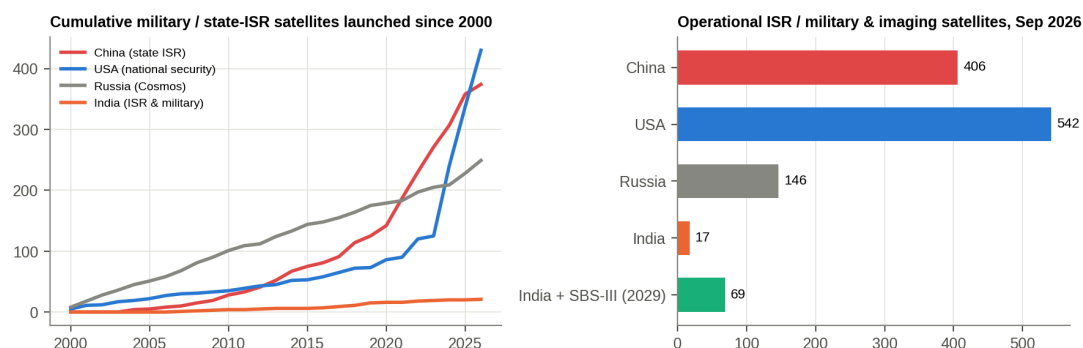


Figure 5.1 : Military and state intelligence satellites: launched since 2000, and working today

What this shows: Left: military and state intelligence satellites launched since 2000. Right: how many are working today.

Grouping satellites by their official programme names gives a transparent lower bound. China operates about **406** state and military imaging, intelligence and technology satellites (164 of them Yaogan military reconnaissance), and has launched about **43 a year** since 2023. The USA has added 305 national-security payloads since 2024. India operates about **17** and launches fewer than one a year. The ratio is about **24 to 1**.

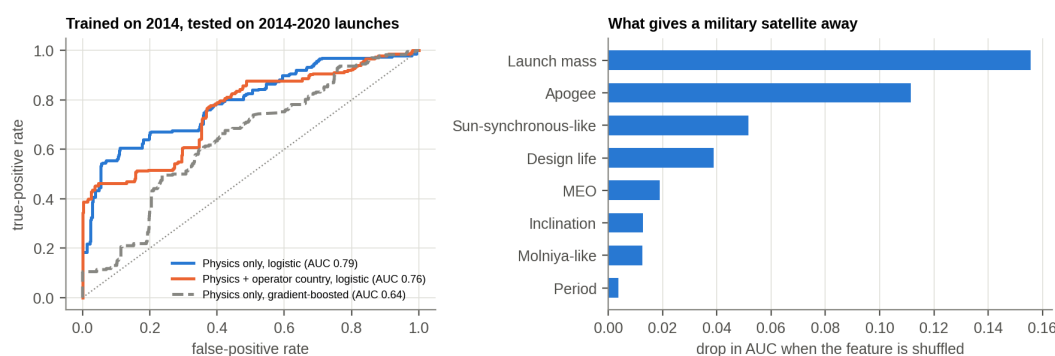
Table 5.1 : Military and state-ISR programme families (name-based lower bound)

Country	Programme family	Operational Sep 2026	Launched since 2014
China	China: Yaogan military reconnaissance	164	149
China	China: other state ISR & tech (Gaofen, TJIS, Shiyang, Shijian, Tianhui, Yunhai)	183	173
China	China: commercial imagers (Jilin-1, Superview)	59	94
USA	USA: national-security payloads (USA-nnn)	417	385
USA	USA: commercial imagers (Planet, Maxar, BlackSky, Capella)	125	581
Russia	Russia: military (Cosmos)	146	125
India	India: ISR & military (RISAT, Cartosat, EOS, EMISAT, GSAT-7)	17	15

India's answer is **SBS-III**: 52 surveillance satellites approved in 2023 (Rs 26,968 crore), 21 built by ISRO and 31 by industry, due by 2029. Against today's gap, SBS-III closes only about **13%**. During Operation Sindoor (May 2025) ten Indian satellites supported the forces round the clock: the need is proven, the capacity is thin.

5.1 Can Physics Reveal a Military Satellite?

Machine learning: orbit, mass and power reveal military use

**Figure 5.2 : Machine learning on the CDM census: predicting military use from orbit, mass and power**

What this shows: Left: how well the model separates military from civil satellites it has never seen (higher curve = better). Right: which physical features give a military satellite away.

Names and registrations can hide a satellite's role; physics cannot. A model trained on the 1,167 satellites of the CDM census (332 military or dual-use) learns military use from orbit, mass, power and design life alone, never the name, purpose or owner. Tested on 1,851 satellites launched **after** the census (UCS 2020), it scores **ROC-AUC 0.79** (0.5 = coin toss). Launch mass and apogee give a satellite away most.

Table 5.2 : Model skill: cross-validation on 2014 against the out-of-time test on later launches

Features	Model	CV ROC-AUC 2014 (mean)	CV SD	Out-of-time ROC-AUC (2020 launches)	Out-of-time PR-AUC	Balanced accuracy @0.5
Physics only	Logistic regression	0.83	0.00	0.79	0.53	0.70
Physics only	Gradient-boosted trees	0.94	0.00	0.64	0.27	0.62
Physics + operator country	Logistic regression	0.84	0.00	0.76	0.55	0.71
Physics + operator country	Gradient-boosted trees	0.94	0.00	0.63	0.27	0.61

The honest test changed the answer. Ordinary cross-validation favoured gradient-boosted trees (ROC-AUC 0.94), but on later satellites they fell to 0.64; the simpler logistic model held at 0.79 and was chosen. Adding the operator's country did not help (0.76): *what* a satellite is says more than *whose* it is. The model flags **83 (5%)** of the later 'civil' or 'commercial' satellites as military-like, led by navigation satellites with encrypted government services: the dual-use blur in numbers, and a triage tool for any new object in the catalogue.

SO WHAT

China operates about 406 state and military imaging and intelligence satellites; India about 17. The gap is not closing on its own. Physics alone flags military satellites (ROC-AUC 0.79 on unseen launches).

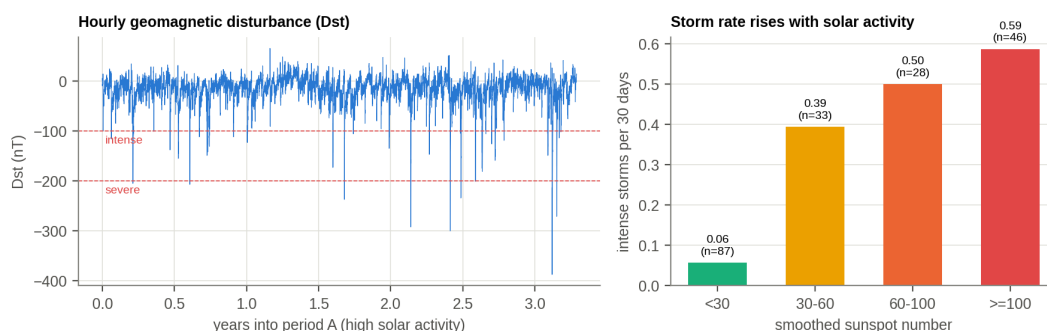
Next: What threat does nature add?

6. THE NATURAL ADVERSARY

PART V - THE NATURAL ADVERSARY

THE STORY SO FAR

Rivals are not the only threat. The Sun sets the weather in space, and it does not negotiate.

The Sun is the one adversary nobody can deter**Figure 6.1 : Geomagnetic storms and solar activity (CDM space-weather data)**

What this shows: Left: an hourly record of magnetic storms (dips below the red lines are dangerous). Right: storms per month at different levels of solar activity.

The CDM records hold 557 storms, 59 of them intense and 13 severe (lowest Dst -387 nT). When the Sun is active (sunspot number 100 or more), the chance of an intense storm in any 30 days is **39%**, against 6% when it is quiet: about **7 times**. A Poisson model gives 1.88 times the storm rate for every 50 more sunspots ($p < 0.001$).

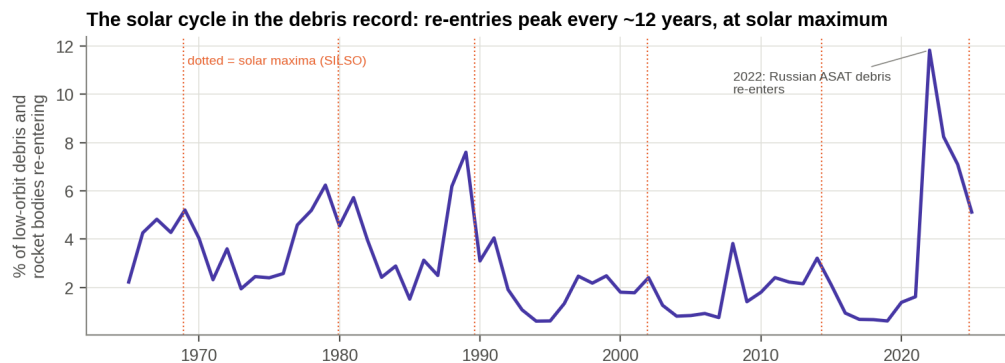


Figure 6.2 : The solar cycle in the debris record: share of low-orbit debris re-entering each year

What this shows: The share of debris falling back to Earth each year. The peaks line up with the Sun's 11-year maximum.

A heated upper atmosphere drags satellites down. A periodogram of re-entry rates finds a **12-year cycle** without being told to look for one: the Sun's cycle. At solar maximum 3.9% of low-orbit debris re-enters each year, against 2.3% at minimum. The same drag killed 38 new Starlink satellites in a moderate storm in February 2022, and the May 2024 'Gannon' superstorm (Dst -412 nT) forced mass orbit corrections. Solar cycle 25 peaked in October 2024 and storms remain frequent through 2026.

SO WHAT

At high solar activity the odds of an intense geomagnetic storm in any month rise about 7-fold. The solar cycle even shows in how fast debris falls.

Next: Why take these risks at all?

7. WHAT EYES IN ORBIT BUY

PART VI - WHAT EYES IN ORBIT BUY

THE STORY SO FAR

So far the story has been about risk. Why do nations accept that risk? Because of what satellites see. The CDM fire data show it on the ground.

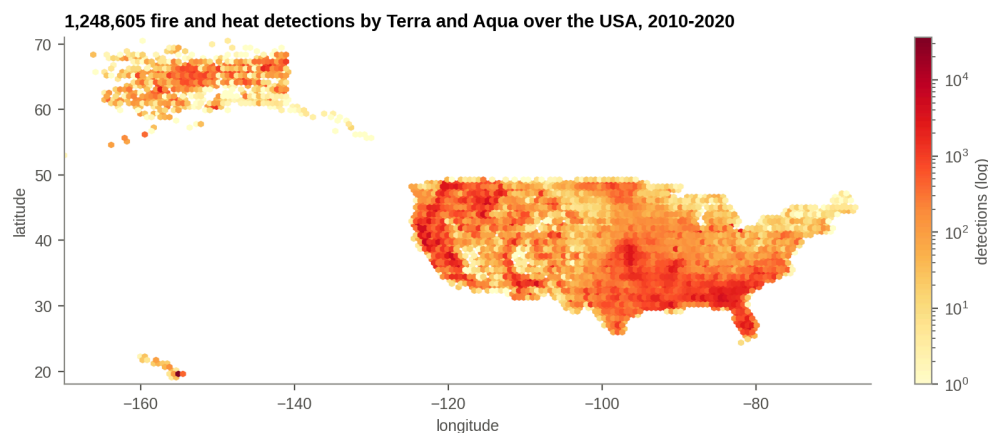


Figure 7.1 : Fire and heat detections by NASA's Terra and Aqua satellites over the USA, 2010-2020 (CDM data)

What this shows: Every fire and heat source the two satellites detected over ten years.

What two satellites deliver: coverage, persistence and early warning (MODIS, 2010-2020)

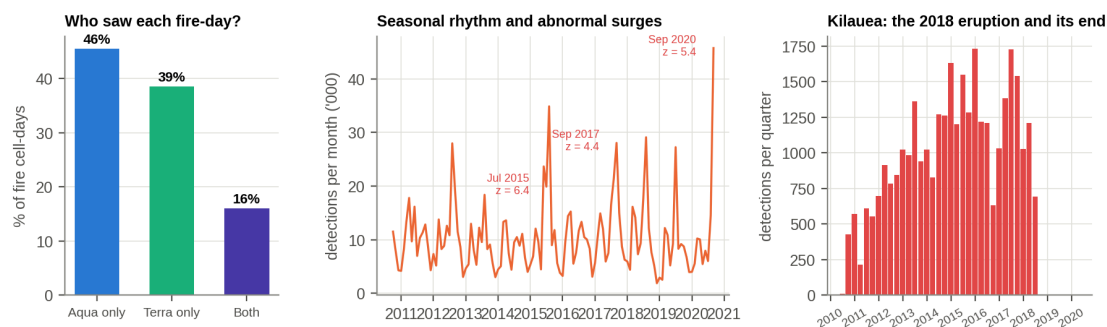


Figure 7.2 : Coverage, persistence and early warning from two satellites

What this shows: Left: which satellite saw each fire-day. Centre: monthly fire detections, with abnormal surges flagged. Right: a volcano watched through its eruption.

Terra passes in the morning and Aqua in the afternoon. Put every detection on a 10-km grid, one cell per day: **84% of fire-days were seen by only one of the two** (46% Aqua only, 38% Terra only). A single satellite would have missed nearly half of what the pair saw. Thermal sensors also see at night (21% of detections). They pick out 659 persistent industrial heat sources, such as steel mills and smelters, and they followed Kilauea's 2018 eruption to its end.

A seasonal baseline flags abnormal months, the logic of any warning system: 11 surges stand out, the largest in Jul 2015 ($z = 6.4$). **For the Services the lesson is direct: persistence comes from numbers.** Revisit, not resolution alone, decides whether a mobilisation, a launcher or a convoy is caught in time.

SO WHAT

Two satellites together saw fires that either one alone would have missed 84% of the time. Numbers in orbit buy coverage, persistence and early warning.

Next: What will orbit look like by 2030?

8. THE TRAJECTORY TO 2030

PART VII - THE TRAJECTORY

THE STORY SO FAR

The past is measured. What comes next?

The trajectory: tens of thousands of satellites by 2030, and a sky many times riskier

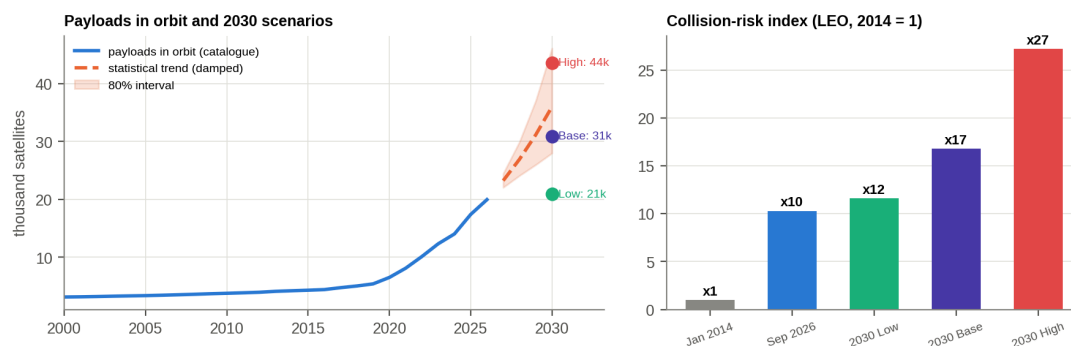


Figure 8.1 : Satellites in orbit to 2030: statistical trend, scenarios and the collision-risk index

What this shows: Left: satellites in orbit and three scenarios for 2030. Right: how collision risk grows under each.

Table 8.1 : Working satellites in 2030 under three deployment scenarios (assumptions stated)

Scenario	Starlink (US)	Kuiper (US)	Guowang (China)	Qianfan (China)	All other satellites	Total 2030
Low	12,000	1,600	800	800	5,701	20,901
Base	15,000	3,236	3,000	3,000	6,639	30,875
High	19,400	3,236	6,500	6,500	7,970	43,606

Two independent routes agree. A damped-trend forecast of satellites in orbit reaches about **36,000** in 2030 (80% range 28,000-46,100); it missed 2022-2026 by only 8.6% when tested on held-back years. A bottom-up build from the announced constellations gives **20,901 / 30,875 / 43,606** working satellites (low / base / high). The collision-risk index rises to about **17 times** the 2014 level in the base case and 27 times in the high case. China's shells at 1,050-1,200 km are the concern: debris there stays for centuries.

SO WHAT

About 31,000 working satellites by 2030 in the base case, and a collision-risk index about 17 times the 2014 level.

Next: How sure are we?

9. INFERENCE: WHAT WE KNOW AND HOW SURE WE ARE

PART VIII - HOW SURE WE ARE

THE STORY SO FAR

The findings are strong claims. Before a commander acts on them, a staff officer asks: are we sure?

Table 9.1 : Every headline finding checked by an independent route

Finding	Result	Independent check	Verdict
Growth of orbit	15x active satellites 2014-2026	UNOOSA registrations track catalogue launches ($r = 0.997$); UCS 2020 census sits on the same curve	Confirmed
Collision risk	index x10 since 2014	Same result with 25 km shells and with objects' mean altitude; the densest shell holds mostly working satellites, the second band mostly debris	Confirmed (relative index)
ISR race	China 406 vs India 17 state ISR/military satellites	Name-based lower bound; excluding commercial imagers the ratio stays above 20:1	Confirmed (lower bound)
Space weather	intense-storm odds x7 at high solar activity	Poisson rate ratio 1.88 per +50 sunspots ($p < 0.001$); Spearman rho -0.57	Confirmed
Solar cycle in debris decay	re-entry cycle of 12 years	Found blind by periodogram; matches the ~11-year solar cycle (SILSO maxima)	Confirmed
Forecast to 2030	~36k payloads (statistical) vs 31k working (base scenario)	Back-test error 8.6% over 2022-2026; two independent routes agree	Moderate confidence
Military-use classifier	out-of-time ROC-AUC 0.79 (trained 2014, tested on 2014-2020 launches)	Cross-validation alone would have picked gradient-boosted trees (CV 0.94); out of time they fell to 0.64, so the simpler model was chosen	Confirmed (out of time)
Close-approach screening	18,355 crossing encounters < 5 km in two 24-hour screenings	Elements used only inside their fresh window (under ~3 days old); co-orbiting and docked pairs removed	Screening grade (~1 km)

Table 9.2 : Key judgements (intelligence estimative language)

How likely	Judgement	Confidence
Almost certain	More than 20,000 satellites will be working by 2030 (even the low scenario).	High
Highly likely	Collision risk in low orbit keeps rising: index 12-27 times 2014 by 2030.	Moderate
Highly likely	China's lead in state ISR satellites over India persists through 2029, even with SBS-III.	High
Likely	At least one intense geomagnetic storm per quarter while sunspots stay above 100 (77%).	Moderate
Realistic possibility	Another debris-creating event (collision, break-up or weapon test) in a crowded shell by 2030.	Moderate
Almost certain	More satellites buy coverage: one satellite misses much of what a constellation sees.	High
Almost certain	Indian satellites pass within 5 km of weapon-test and collision debris every day (10 in one day).	Moderate

The judgements use the estimative language of intelligence assessments. Confidence reflects the data: counts from the catalogue are high-confidence; the 2030 scenarios depend on company and state deployment plans and are moderate.

SO WHAT

Every headline finding survives an independent check.

Next: What does it mean for India and its Armed Forces?

10. WHAT IT MEANS FOR THE WORLD, INDIA AND THE ARMED FORCES

PART IX - WHAT IT MEANS

THE STORY SO FAR

The evidence holds. What does it mean for the world, for India, and for the Army, Navy and Air Force?

For the world: orbit is becoming a commons run by a few operators. A single company's decisions now shape the collision risk for everyone, and commercial constellations have become military assets in war. Norms on debris and weapon tests lag far behind the traffic.

Table 10.1 : India against China and the USA

Measure	India	China	USA
Active satellites, Sep 2026	78	1,531	12,959
Active satellites, Jan 2014 (CDM)	33	117	502
Military / state-ISR satellites operational	17	406	542
Orbital launches in 2025	4	89	172
GEO satellites in the 40-110 E arc (2020)	24	32	-

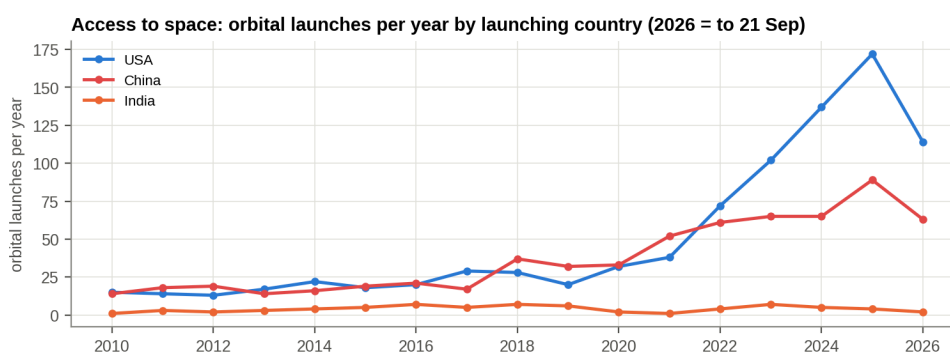


Figure 10.1 : Access to space: rocket launches per year from US, Chinese and Indian soil

What this shows: Rocket launches per year from US, Chinese and Indian soil.

For India: national dependence on space grows (banking, navigation, communications, disaster warning), while India's share of working satellites has fallen to 0.5% and it launched 4 rockets in 2025 against China's 89. Over 2021-25 China averaged about 16 times India's launch rate. Access to space is the bottleneck.

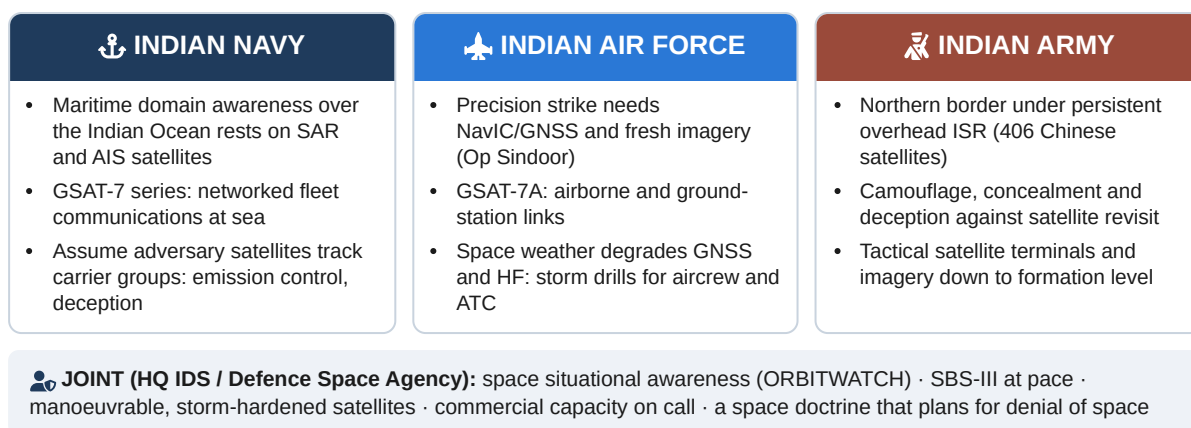


Figure 10.2 : Impact on the Indian Navy, Air Force and Army, and the joint response

For the Armed Forces: the adversary can now watch continuously; our own satellites face collisions, storms and counterspace threats; and operations increasingly assume space support that may be denied. Plans must assume **being watched** and **losing some space services**.

SO WHAT

India depends on space more every year while holding a shrinking share of it. That is the strategic problem.

Next: What should India do?

11. WAY FORWARD

PART X - WHAT WE SHOULD DO

THE STORY SO FAR

The implications are clear. The last question is the one a commander always asks: what do we do, who does it, and by when?

Table 11.1 : Seven staffed actions, traced to the evidence

Action	Lead	Timeline	Measure of effectiveness
Stand up ORBITWATCH: daily close-approach screening (demonstrated in 4.1) and the 7-indicator warning matrix from open, ISRO NETRA and commercial data	DSA with ISRO (NETRA)	4 months	An actionable conjunction or threat warning acted upon
Accelerate SBS-III and buy commercial imagery and radar capacity for persistent revisit of the northern borders and IOR	DSA, NSIL, IN-SPACE, industry	2026-2029	Revisit time over priority areas
Resilience by design: manoeuvre capability, collision-avoidance fuel and storm hardening for every new military satellite	DSA, ISRO, DRDO	Contracts from 2027	Share of fleet able to manoeuvre
Space-weather cell: storm warnings to all Services; drills for degraded GNSS and HF communications	HQ IDS, IAF Met, ISRO	6 months	Drill conducted each quarter
Rapid reconstitution: SSLV launch slots and spare satellites on standby	ISRO, NSIL, DSA	2-3 years	Days to replace a lost satellite
Protect GEO slots and ITU filings in the 40-110 E arc; interference monitoring	DoT, ISRO, DSA	Ongoing	Filings secured; interference cases resolved

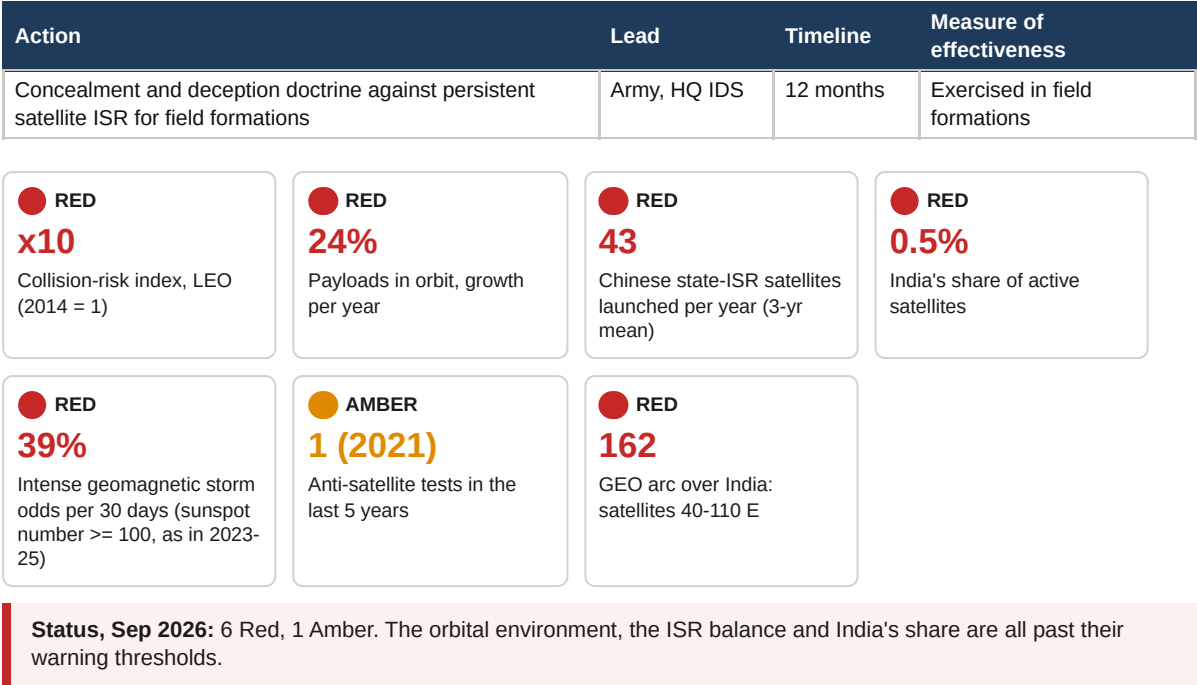


Figure 11.1 : ORBITWATCH: the seven-indicator warning matrix at September 2026



Figure 11.2 : Three-phase plan

SO WHAT

Seven actions, each with a lead, a timeline and a test of success.

12. CONCLUSION

EPILOGUE

THE STORY SO FAR

We end where we began, with the commander's questions, now answered with evidence.

✖

Orbit is crowded. 15 times more working satellites than in 2014; about 31,000 by 2030.

🚩

Orbit is concentrated. One country operates 76% of working satellites; India 0.5%.

☄

Orbit is dangerous. Collision risk about x10; high-altitude weapon debris lasts decades.



Orbit is contested. China operates about 406 state ISR satellites; India about 17.



Numbers and resilience win. More satellites buy coverage; manoeuvre, hardening and awareness keep them alive.

Figure 12.1 : Five takeaways

The high ground has become a crowd. In twelve years orbit went from about a thousand working satellites to seventeen thousand, owned mostly by one country and one company, circling in shells where collision risk has multiplied and where weapon debris can linger for decades. China is filling it with eyes; the Sun periodically lashes it. India, which depends on space more every year, holds a shrinking share of it. The remedy is not a single prestige satellite but numbers, resilience and awareness: SBS-III at pace, satellites that can manoeuvre and survive, and a standing ORBITWATCH so that India always knows what is overhead.

ANNEX A - DATA ANALYTICS IN PLAIN WORDS

Table A.1 : Glossary of techniques and terms

Term	What it means, in plain words
Satellite catalogue (SATCAT)	The public list of every tracked object in Earth orbit: satellites, rocket bodies and debris, with launch and re-entry dates and orbit heights.
LEO / MEO / GEO	Low Earth orbit (below 2,000 km, most satellites), medium orbit (navigation satellites) and geostationary orbit (36,000 km above the equator, where a satellite appears fixed in the sky).
Mega-constellation	Thousands of small satellites flying together to give continuous global coverage, such as Starlink.
Debris	Broken pieces of satellites and rockets: from explosions, collisions and weapon tests. At 7-8 km/s a 1 cm piece hits like a hand grenade.
ASAT test	Anti-satellite weapon test: a missile destroys a satellite in orbit, creating a cloud of debris.
Collision-risk index	A relative measure of how likely collisions are: it grows with the square of the number of objects at each height. $2014 = 1$.
HHI (concentration index)	Sum of squared market shares, 0-10,000. Above 2,500 means a few players dominate.
Change-point detection	A method that finds the date when a trend shifted, without being told where to look.
Dst index	A measure of how much a magnetic storm disturbs the Earth's field. Below -100 nT is an intense storm.
Sunspot number	A count of dark spots on the Sun; high counts mean an active Sun and more storms.
Periodogram	A tool that finds repeating cycles in a time series, like finding the beat in music.
Poisson regression	A model for counts (e.g. storms per month) that measures how a factor raises the rate.
Constellation premium	What extra satellites add: the share of events that one satellite alone would have missed.
Exponential smoothing (Holt)	A forecasting method that follows the recent trend and lets it flatten over time.
Back-test	Hiding the last few years, forecasting them, and checking the error: a test of the forecast method.
ISR	Intelligence, surveillance and reconnaissance: satellites that image, listen and watch.
ORBITWATCH	The seven-indicator warning matrix proposed in this study, updated from open data.
SGP4	The standard mathematical model that turns published orbital elements into a satellite's position at any time.
Close approach (conjunction)	Two objects passing close to each other; screening flags passes under 5 km for closer study.
Out-of-time test	Training a model on older data and testing it on later data it has never seen: the honest test of a forecast.
ROC-AUC	A model's skill at ranking positives above negatives: 0.5 is a coin toss, 1.0 is perfect.
SBS-III	India's Space-Based Surveillance phase III: 52 military surveillance satellites approved in 2023, due by 2029.

ANNEX B - TECHNICAL SUMMARY FOR ASSESSORS

Every technique used, the question it answers and the key result. All results are regenerated from the raw data by one command (analysis/run_all.py), and two consecutive runs give identical outputs.

Table B.1 : Techniques, questions and results

Technique	Commander's question	Key result	Chapter
Validation and cleaning: Excel-date conversion, impossible periods/inclinations removed, code look-ups, duplicate removal	Is the evidence sound?	6 impossible periods and 1 impossible inclination corrected; 1 duplicate fire removed	Ch 1
Year-end population rebuilt from launch and re-entry dates; three independent censuses	How crowded is orbit?	1,167 to 17,062 working satellites	Ch 2
Change-point detection (PELT); compound growth rates; cross-check with UNOOSA	When did growth change gear?	Break 2021; 4.1% to 46.7% a year; $r = 0.997$	Ch 2
Herfindahl-Hirschman concentration index	Is control of orbit spreading or concentrating?	HHI 2,171 to 5,879	Ch 3
Altitude-shell spatial density; kinetic-gas collision-risk index ($\sum n^2 / V$)	How dangerous has orbit become?	Index $\times 10.3$	Ch 4
Survival curves of debris by event	How long does weapon debris last?	FY-1C 65.5% remaining; Shakti 0%	Ch 4
Name-based programme families (transparent lower bound)	Who is building military eyes?	China 406 vs India 17	Ch 5
Storm event detection (Dst thresholds); Poisson regression; Spearman correlation	Does solar activity raise storm risk?	Rate ratio 1.88 per +50 sunspots; odds $\times 6.8$	Ch 6
Periodogram of de-trended re-entry rate	Does the solar cycle show in debris decay?	12.2-year cycle	Ch 6
Grid-day matching of detections (0.1 deg); seasonal baseline z-scores	What does a constellation add?	84% of fire-days seen by one satellite only	Ch 7
Holt damped-trend smoothing with 2,000 simulations and back-test; bottom-up scenarios	What comes by 2030?	36,000 (trend); 30,875 (base); back-test 8.6%	Ch 8
Logistic regression and gradient-boosted trees; repeated stratified CV; out-of-time test on UCS 2020 launches; permutation importance	Can physics reveal a military satellite?	Out-of-time ROC-AUC 0.793 (trees 0.644); 83 dual-use flags	Ch 5.1
SGP4 propagation every 10 s for 24 h; k-d tree pair screening (80 km); 1-s refinement and linearised closest approach	How often do satellites nearly collide?	18,355 encounters < 5 km in two screenings; 10 Indian vs debris	Ch 4.1
Indicators and warnings matrix (ORBITWATCH)	What should be watched?	6 Red, 1 Amber	Ch 11

Table B.2 : Data cleaning log

Dataset	Step	Count
CDM satellites	Rows loaded	1,167
CDM satellites	Launch dates converted from Excel serial numbers (1 missing)	1
CDM satellites	Impossible orbital periods (< 80 min) set to missing	6
CDM satellites	Impossible inclinations (> 180 deg) set to missing	1
CDM satellites	Rows after cleaning	1,167
UCS 2020	Active satellites	2,666
SATCAT	Objects catalogued (1957 - Sep 2026)	70,793
SATCAT	Earth-orbiting objects kept	70,366
SATCAT	Objects without orbital elements (kept; excluded from altitude work)	1,657
Space weather	Hourly Dst values (3 periods)	139,872
Space weather	Monthly smoothed sunspot values	192
MODIS fires	Detections loaded (archive + near real time)	1,248,606
MODIS fires	Duplicate detections removed	1
MODIS fires	Rows after cleaning	1,248,605

ANNEX C - SOFTWARE USED AND SCREENSHOTS

Stack: Python 3.11 (pandas, NumPy, SciPy, statsmodels, scikit-learn, ruptures, sgp4, Matplotlib) for analysis; headless Chromium for the booklet; Microsoft Power BI Desktop for the dashboard (.pbix) built on the exported tables (FactCatalogue, FactSatellites2014, FactStormDaily, FactFiresMonthly, DimDate and the analysis tables).

```
Terminal - Theme 6.1 - full pipeline run (raw data to booklet)

$ python analysis/run_all.py
Stage 1: preprocessing
Stage 2: growth
  fig -> f2_1_population
  fig -> f2_2_censuses
  fig -> f2_3_launch_rate
[3 rows x 3 columns]
Stage 3: actors
  fig -> f3_1_actors
  fig -> f3_2_users_purpose
[8 rows x 6 columns]
[8 rows x 4 columns]
Stage 4: congestion
  fig -> f4_1_leo_shells
  fig -> f4_2_geo_arc
  fig -> f4_3_asat_debris
Stage 5: contest
  fig -> f5_1_isr_race
[7 rows x 15 columns]
Stage 6: space weather
  fig -> f6_1_storms
  fig -> f6_2_reentry_cycle
Stage 7: Earth-observation value
  fig -> f7_1_eo_value
  fig -> f7_2_fire_map
Stage 8: forecast
  fig -> f8_1_forecast
Stage 11: military-use classifier
  fig -> f11_1_ml_military
Stage 12: conjunction screening
  fig -> f12_1_conjunctions
Stage 9: India, robustness, ORBITWATCH
  fig -> f9_1_launch_capacity
Stage 10: Power BI export
FactCatalogue.csv: 70,366 rows
FactSatellites2014.csv: 1,167 rows
FactStormDaily.csv: 5,828 rows
FactFiresMonthly.csv: 484 rows
DimDate.csv: 25,567 rows
... 48 figures and tables written
$
```

Figure C.1 : Screenshot: the complete pipeline run, raw data to every chart and table

```
analysis/04_congestion.py - altitude shells and the collision-risk index

30 # ----- 1-2. LEO shells and the collision-risk index
31 W = 25 # shell thickness, km
32 edges = np.arange(200, 2000 + W, W)
33 vol = 4 / 3 * np.pi * ((EARTH_R + edges[1:]) ** 3 - (EARTH_R + edges[:-1]) ** 3) # km^3 per shell
34
35
36 def shells(df):
37     leo = df[(df.APOGEE < 2000) & df.alt_mean.notna()]
38     out = {}
39     for t in ["Payload", "Debris", "Rocket body"]:
40         out[t] = np.histogram(leo[leo.type.eq(t)].alt_mean, edges)[0]
41     out["All"] = np.histogram(leo.alt_mean, edges)[0]
42     out["Active"] = np.histogram(leo[leo.operational].alt_mean, edges)[0] if "operational" in leo else 0
43     return pd.DataFrame(out, index=edges[:-1])
44
45
46 s14, s26 = shells(on_orbit(c, SNAPSHOT_CDM)), shells(on_orbit(c, CATALOGUE_DATE))
47 risk = lambda n: float((n ** 2 / vol).sum())
```

Figure C.2 : Screenshot: code excerpt, altitude shells and the collision-risk index

[Insert screenshot: Power BI page 1 - Orbit Growth]

Figure C.3 : Screenshot: Power BI dashboard, page 1 - Orbit Growth (working satellites, launches and population by year)

[Insert screenshot: Power BI page 2 - Owners and Contest]

Figure C.4 : Screenshot: Power BI dashboard, page 2 - Owners and Contest (country shares and ISR families)

[Insert screenshot: Power BI page 3 - Congestion]

Figure C.5 : Screenshot: Power BI dashboard, page 3 - Congestion (altitude shells, GEO arc and debris events)

[Insert screenshot: Power BI page 4 - Space Weather and Warning]

Figure C.6 : Screenshot: Power BI dashboard, page 4 - Space Weather and Warning (storms, re-entries and ORBITWATCH)

ANNEX D - REFERENCES AND DATA SOURCES

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